

Brady Deschamps

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Summary

Computer Science BS/MS candidate at Arizona State University with experience building automated ML evaluation pipelines, Python data pipelines, and API-driven software. Skilled in designing build-run-evaluate-iterate workflows, computer vision, software testing, and cloud or high-performance computing environments, with cybersecurity research experience focused on attacker behavior. Comfortable collaborating across technical teams and documenting reliable systems.

Education

Arizona State University

Aug 2024 - May 2027

BS Computer Science (GPA: 4.0)

- **Achievements:** Barrett, The Honors College, Tillman LTA Scholar, Grand Challenges Scholars Program
- **Coursework:** Data Structures and Algorithms, Database Management, Software Engineering

Experience

SMBC | Information Technology Intern

Jun 2026 - Aug 2026

- Developed a full-stack port-tracking tool using React, Node, SQL, and the Nlyte NGage API, reducing manual tracking effort and improving inventory data accuracy for datacenter operations
- Monitored, documented, and resolved infrastructure tickets across two enterprise datacenter locations for one of the largest foreign banking institutions in the U.S., ensuring operational continuity and timely issue resolution
- Installed and configured new network devices bridging on-premises datacenter infrastructure with cloud service providers, collaborating with the network team to integrate hardware into hybrid cloud architecture

The Luminosity Lab | Software Engineer

Feb 2026 - Present

- Built an automated ML inference pipeline on ASU's Sol supercomputer, orchestrating three sequential deep learning models with SLURM job arrays and Apptainer containers, using a feedback loop on output-scoring metrics to grade runs and direct subsequent parameters
- Engineered a multithreaded computer vision detection system with PyTorch and parallel processing techniques to optimize performance and reduce inference latency for image recognition tasks
- Architected and deployed a Python-based end-to-end data pipeline integrating NASA JPL datasets with the OrbitalScape platform, enabling real-time aerospace data visualization and analysis

Pivotal Energy Solutions | Software Engineer Intern (Part-time)

Apr 2024 - Jun 2026

- Refactored Python serializer validation logic into a JSON-based REST API, enabling language-agnostic data verification and improved system scalability
- Increased unit test code coverage by 90% across 20+ interconnected modules by writing comprehensive test suites in Python, improving reliability and maintainability

Arizona State University | Cybersecurity Research Assistant

Nov 2024 - Aug 2025

- Designed and deployed an intentionally vulnerable system to capture live attacker behavior, building behavioral intrusion profiles used to inform development of a personalized honeypot AI model

School of Computing and Augmented Intelligence | SCAI Peer Mentor

Aug 2025 - Present

- Guided students in SCAI on academic and professional development, providing technical mentorship in CS coursework and career preparation

Projects

Datatable Advanced Query | ICMange

- Implemented a lexer and parser for a SQL-like search syntax supporting boolean operators, comparison operators, and complex expressions
- Integrated with the Django ORM to translate the parsed abstract syntax tree into database queries

OrbitalScape | OrbitalScape.space

2025 - Present

- Applying machine learning techniques to space weather analysis within the OrbitalScape platform, contributing to predictive modeling efforts for orbital and aerospace applications, built on a production Python/pandas data pipeline connecting NASA JPL data sources to the platform

Skills

- **Programming Languages:** Python, TypeScript, JavaScript, Java, C++, SQL
- **Frameworks and Libraries:** React, Django, PyTorch
- **Engineering and Development:** APIs, Object Oriented Programming, Software Testing, Machine Learning Model Training, CI/CD
- **Cloud and Infrastructure:** Datacenter Operations, Docker, Cloud Platforms
- **Version Control and Collaboration:** Git, GitHub